International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Examination, Autumn 2022

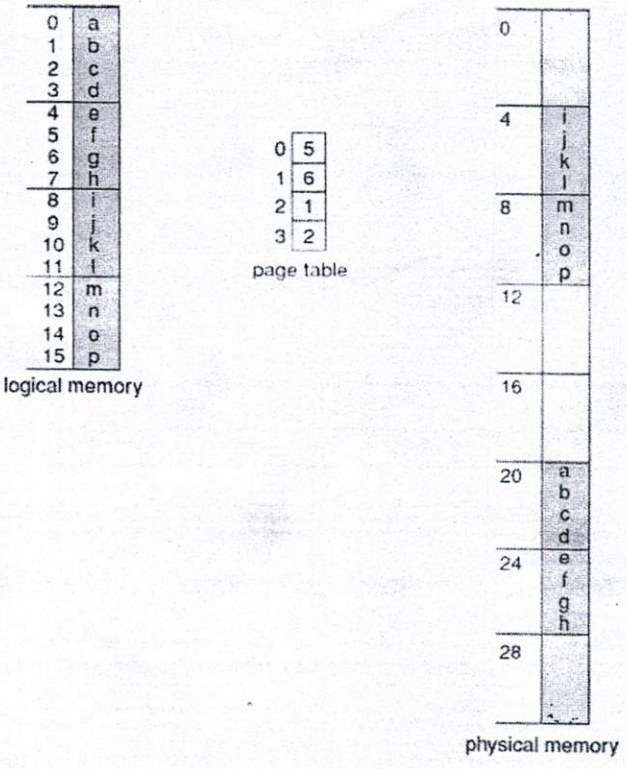
Course Code: CSE 3631 **Course Title: Operating Systems**

Time: 2 Hours 30 Minutes Total Marks: 50

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

		CLO	
1. a)	Consider the following resource-allocation policy. Requests for and releases of resources are allowed at any time. If a request for resources cannot be satisfied because the resources are not available, then we check any threads that are blocked waiting for resources. If a blocked thread has the desired resources, then these resources are taken away from it and are given to the requesting thread. The vector of resources for which the blocked thread is waiting is increased to include the resources that were taken away. For example, a system has three resource types, and the vector Available is initialized to (4, 2, 2). If thread T0 asks for (2,2,1), it gets them. If T1 asks for (1,0,1), it gets them. Then, if T0 asks for (0,0,1), it is blocked (resource not available). If T2 now asks for (2,0,0), it gets the available one (1,0,0), as well as one that was allocated to T0 (since T0 is blocked). T0's Allocation vector goes down to (1,2,1), and its Need vector goes up to (1,0,1). 1. Can deadlock occur? If you answer "yes," give an example. If you answer "no," specify which necessary condition cannot occur. 2. Can indefinite blocking occur? Explain your answer.	CLO2	05
	Or,		
	Describe different recovery techniques from deadlock.	CLO2	5
b)	Consider the following snapshot of a system: *Allocation* / *Max* / *Available* A B C D / A B C D / A B C D T_0 3 1 4 1 / 6 4 7 3 / 2 2 2 4 T_1 2 1 0 2 / 4 2 3 2 T_2 2 4 1 3 / 2 5 3 3 T_3 4 1 1 0 / 6 3 3 2 T_4 2 2 2 1 / 5 6 7 5 Decide whether the system is in a safe state or unsafe state by using Banker's algorithm.	CLO2	05
	Or,		
	Define segmentation. Explain the user and logical view of segmentation.	CLO2	5

2. a)	Consider the paging example of the following figure for a 32-byte memory with 4-byte pages.  Now map logical addresses 0, 4,11 and 15 to physical addresses.	CLO3	05
b)	Explain the paging hardware with TLB.	CLO3	03
c)	Consider a paging system with the page table stored in memory. 1. If a memory reference takes 50 nanoseconds, how long does a paged memory reference take? 2. If we add TLBs, and if 75 percent of all page-table references are found in the TLBs, what is the effective memory reference time? (Assume that finding a page-table entry in the TLBs takes 2 nanoseconds, if the entry is present.)	CLO3	02
	Group B		
3. a)	Analyze thrashing effect in the paging system.	CLO3	04
	Or,		
	What is file allocation method? Describe Linked Allocation and Indexed Allocation method.	CLO3	04
b)	I. How many page faults occur for the LRU page replacement algorithm for the following reference string with four page frames? 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2. II. What is the minimum number of page faults for an optimal page replacement strategy for the reference string in part I with four page frames?	CLO3	06
	Or,		
	Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4 page frames. Find number of page faults occur for the following algorithms: 1. FIFO Algorithm 2. Optimal Algorithm 3. LRU Algorithm	CLO3	06
4. a)	In spite of limited physical memory, we can execute a program whose logical memory size is greater than physical memory of the system. How is it possible?	CLO3	02

b)	Consider a file currently consisting of 100 blocks. Assume that the file-control block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. 1. The block is removed from the beginning. 2. The block is removed from the middle. 3. The block is removed from the end	CLO3	05
c)	Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.	CLO3	03
5. a)	Callie wants to send the message M = 13 to Alice. Using Alice's public ((n, e) = (33, 3)) and private keys ((n, d) = (33, 7)). calculate the cipher text C, and the value for R when Alice recovers the message	CLO3	03
b)	How does Trojan Horse destroy a computer system?	CLO3	03
c)	What is the goal of protection? What do you know about firewall to protect system?	CLO3	04